

MATH 1342 – Elementary Statistics – Week 2: Sampling, Experiments and Observational Studies

Sep 1 – 3 | 1.2 Sampling from a Population • 1.3 Experiments and Observational Studies | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 7. Apply ideas of appropriate sampling techniques and experimental design to data production.	1. Distinguish a population from a sample and describe statistical inference. (CO1, CO7) 2. Identify sampling bias in a described study, classify the sampling method (random, convenience, volunteer), and judge whether the results can be generalized. (CO7) 3. Recognize bias from question wording, question context, and inaccurate responses. (CO7) 4. Classify a study as observational or experimental, and identify a plausible confounding variable. (CO7) 5. Explain how random assignment, a control group, and blinding support cause-and-effect conclusions, and distinguish random sampling (generalize) from random assignment (causation). (CO7)	e-Text: 1.2 Sampling from a Population	MO1, MO2, MO3	Participation: Variables in Statistics (Kahoot!) – Tue: review of Week 1	MO1	Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 1.2 Sampling from a Population	MO1, MO2, MO3	Participation: Sampling – spot the bias in sample scenarios	MO2, MO3		
		Slides: 1.2 Sampling from a Population (includes sampling video)	MO1, MO2, MO3	Participation: Sampling Methods (Kahoot!) – Thu: sampling review	MO2, MO3	Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)	MO4, MO5
		e-Text: 1.3 Experiments and Observational Studies	MO4, MO5	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment	MO4, MO5		
		Interactive Lecture: 1.3 Experiments and Observational Studies	MO4, MO5				
		Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	MO4, MO5			Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3	All